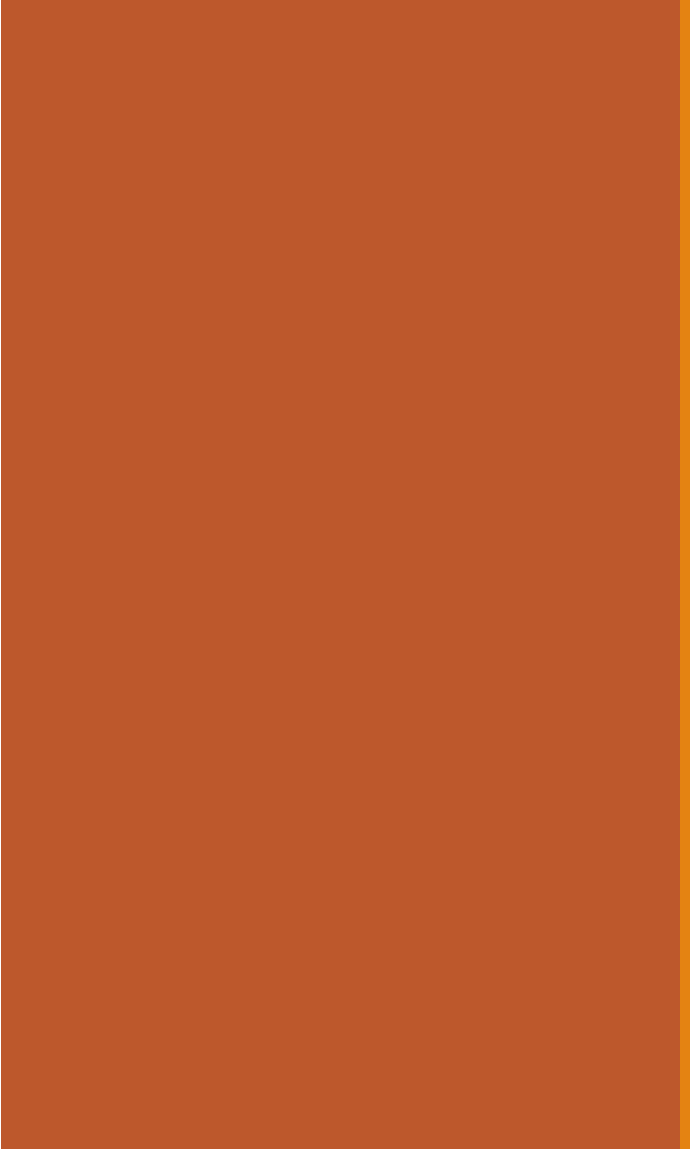


19: Sampling and the Bootstrap

Jerry Cain
May 13th, 2024



[Lecture Discussion on Ed](#)



Sampling definitions

Motivating example

You want to know the true mean and variance of happiness in Bhutan.

- You can't ask everyone ^{all of} very easily, so..
- You poll 200 random people.
- Your poll data looks like this:

Happiness = {72, 85, 79, 91, 68, ..., 71}

average of these 200 numbers is

- The mean of all these numbers is 83. *←*

Is this the **true average happiness** of all Bhutanese people? *Is it close? How often?*



Population

assume for simplicity
that population size is $N=10,000$
(true population size is $\sim 800,000$,
though)



This is a **population**.

Sample



A **sample** is selected from a population.

Sample



We hope and generally assume that, provided the sample was taken correctly and can be assumed to be reliable, it represents the full population very well.

A **sample** is selected from a population.

Reasonable Questions Starting Out

e.g. interview, analyze, inquire, etc.

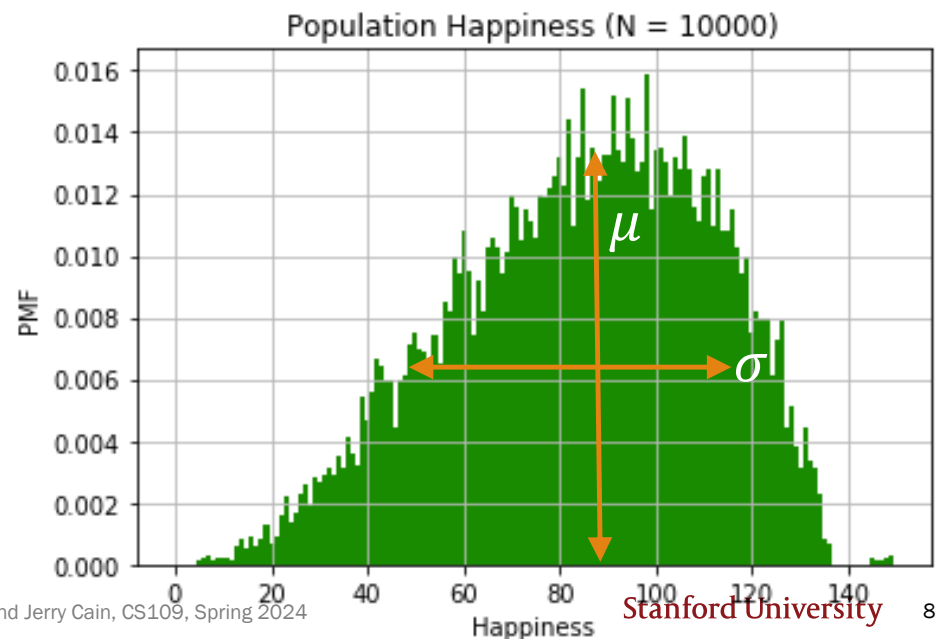
1. In situations where we can't observe the entire population, what can we safely infer by polling a sample drawn from that population?
2. How large does your sample need to be before your conclusions become trustworthy, and how do we express our confidence in those conclusions.
3. Are there alternative ways to infer population statistics without polling entire populations? Sure, though we won't really discuss anything not covered in today's lecture.

A sample, mathematically

Consider n random variables $X_1, X_2, \dots, X_n$.

The sequence $X_1, X_2, \dots, X_n$ is a **sample** from distribution F if:

- X_i are all independent and identically distributed (iid)
- X_i all have same distribution function F (the **underlying distribution**), where $E[X_i] = \mu$, $\text{Var}(X_i) = \sigma^2$



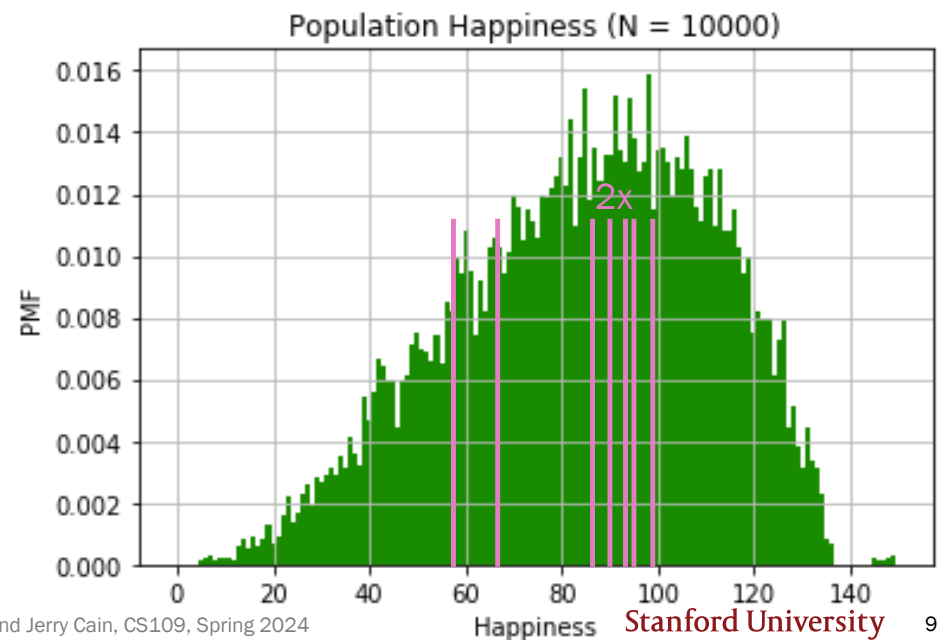
A sample, mathematically

A sample of **size** 8:

$(X_1, X_2, X_3, X_4, X_5, X_6, X_7, X_8)$

The **realization** of a sample of size 8:

$(59, 87, 94, 99, 87, 78, 69, 91)$



A single sample



A happy
Bhutanese person

If we had a distribution F of our entire population, we could compute exact statistics about happiness.

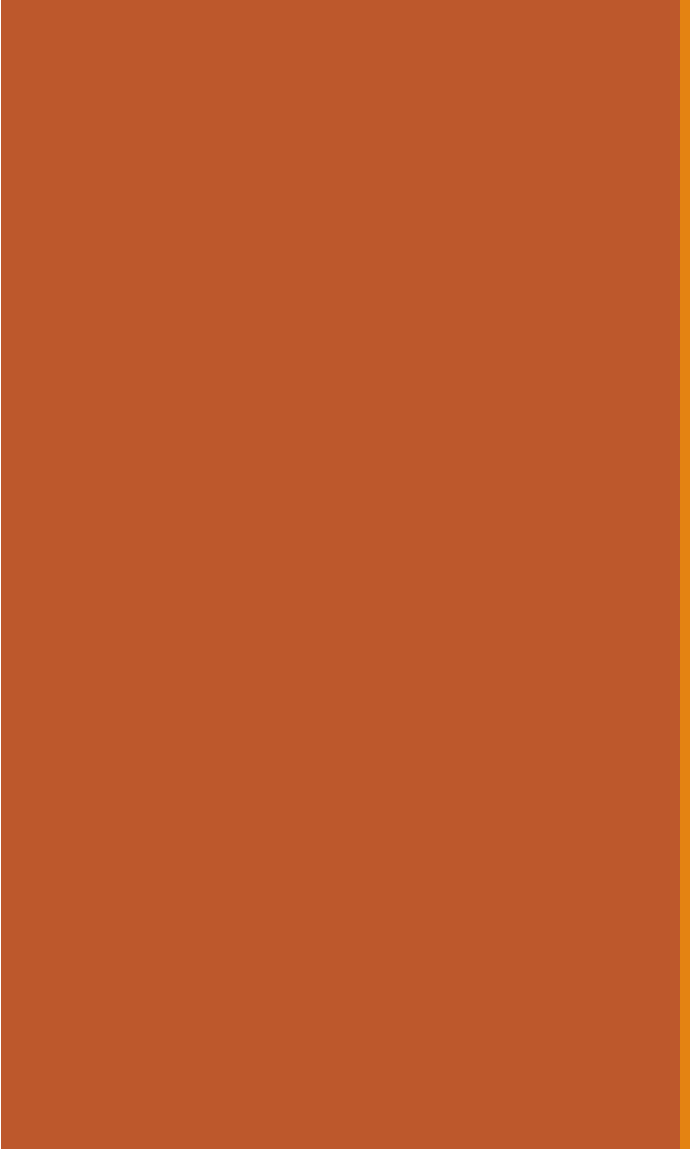
the distributions presented on the previous slides gives the impression we know the full population distribution F .

In general, we don't have that, which is why we need to sample.

But we only have 200 people—or rather, a sample.

Assuming we have just a single sample,

- how do we report **estimated** statistics?
 - We're careful to call them **estimated** mean and **estimated** variance, since they're based on a sample
- How do we report confidence intervals on these estimates? *these confidence intervals are often called standard errors.*
- How do we perform something called **hypothesis testing**? Oh, and what is it? *} we'll cover this next time.*



Unbiased estimators

A single sample



A happy
Bhutanese person

If we had a distribution F of our entire population, we could compute exact statistics about happiness. *But again, we don't usually have this. If we did, we wouldn't need sampling theory.*

But we only have a sample of happiness scores from 200 people.

These population-level statistics are unknown:

- μ , the **population mean**
- σ^2 , the **population variance**

A single sample



A happy
Bhutanese person

If we had a distribution F of our entire population, we could compute exact statistics about happiness.

But we only have a sample of happiness scores from 200 people.

- Using this sample, what is our best guess estimate of the **population mean** and the **population variance**?
- How exactly do we define best guess estimate?

Estimating the population mean

best estimate and best guess
estimate are the same
thing.



1. What is our best estimate of μ , the **mean happiness** of Bhutanese people?

all

If we only have $(X_1, X_2, \dots, X_n)$:

The best estimate of μ is the **sample mean**:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

$\bar{X}$ is an unbiased estimator of the population mean μ .

$$E[\bar{X}] = \mu$$

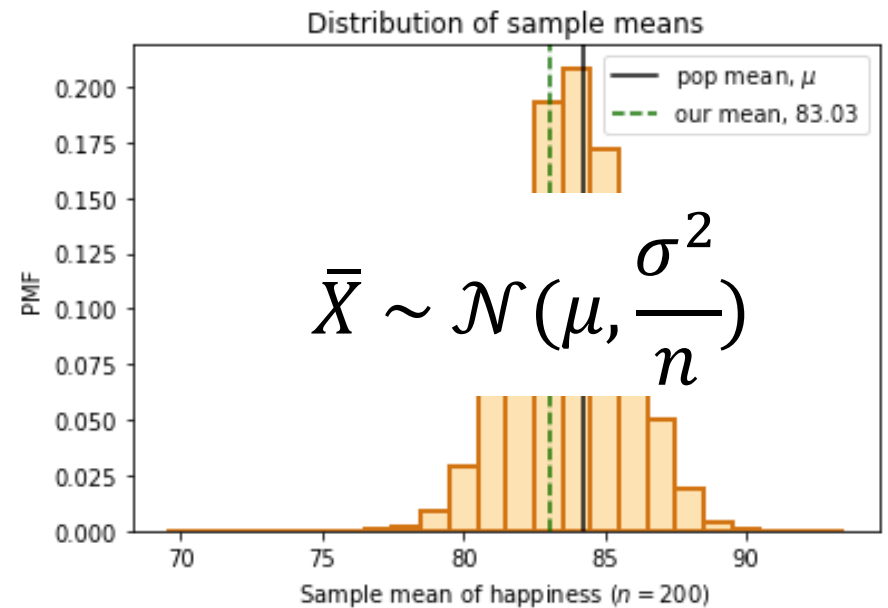
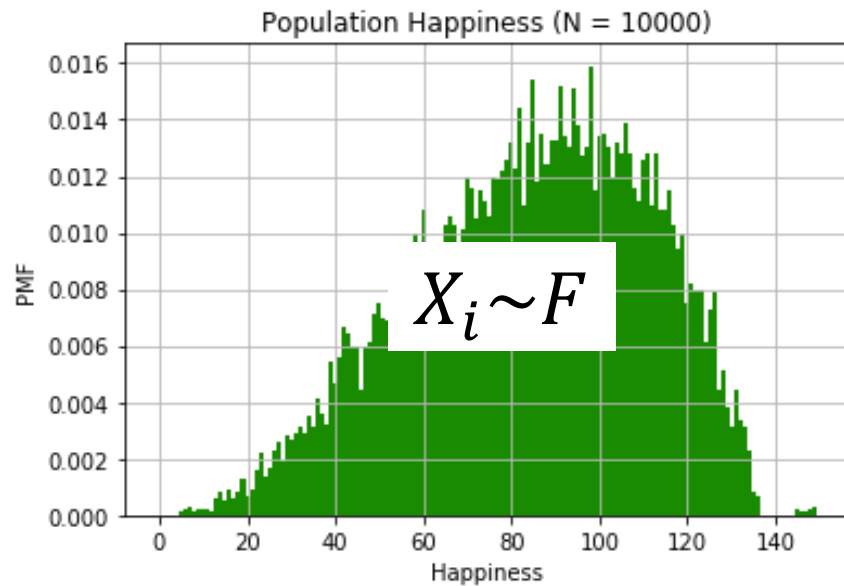
Intuition: By the CLT, $\bar{X} \sim \mathcal{N}(\mu, \frac{\sigma^2}{n})$



If we could take *multiple* samples of size n :

1. For each sample, compute sample mean
2. On average, we would get the population mean

Sample mean



Even if we can't report μ , we can report our sample mean of 83, which is an unbiased estimate of μ .

Estimating the population variance



2. What is σ^2 , the **variance of happiness** of Bhutanese people?

If we knew the entire population $(x_1, x_2, \dots, x_N)$:

population variance

$$\sigma^2 = E[(X - \mu)^2] = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

population mean

If we only have one sample: $(X_1, X_2, \dots, X_n)$:

sample variance

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

sample mean

two observations:

1.) denominator is $n-1$ instead of n . why not n ?

2.) this estimated variance is defined in terms of another estimate — the sample mean.

Intuition about the sample variance, S^2

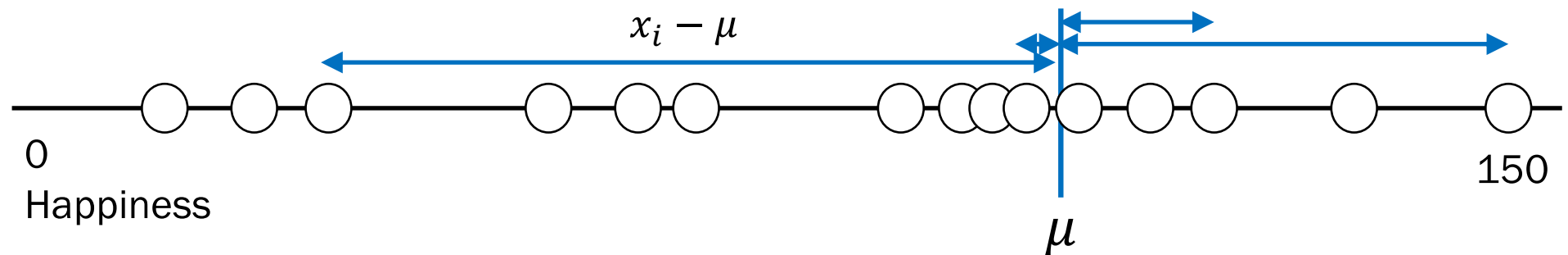
Actual, σ^2

population
variance

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

population mean

*you have perfect information
if you have access
to the full population
distribution.*



Population size, N

Calculating population statistics exactly
requires us knowing all N datapoints.

Intuition about the sample variance, S^2

Actual, σ^2

population variance

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

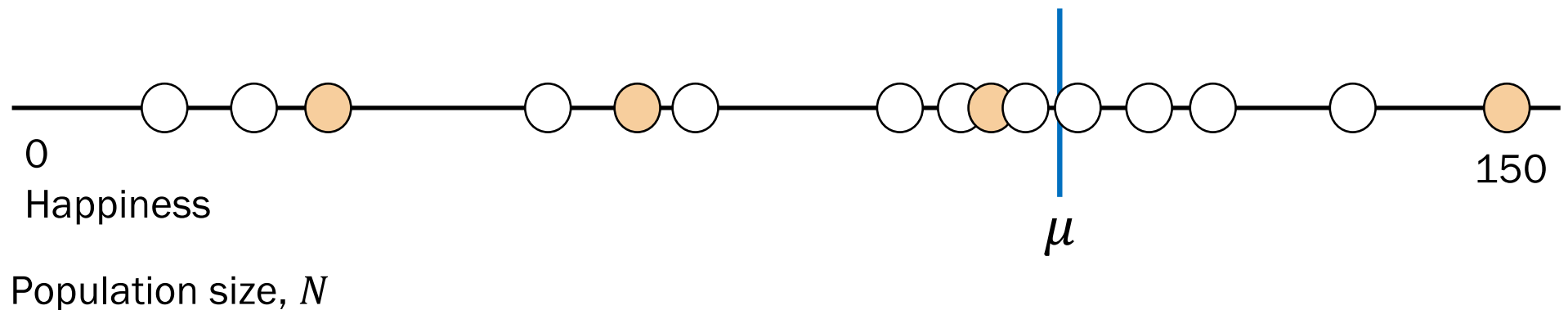
population mean

Estimate, S^2

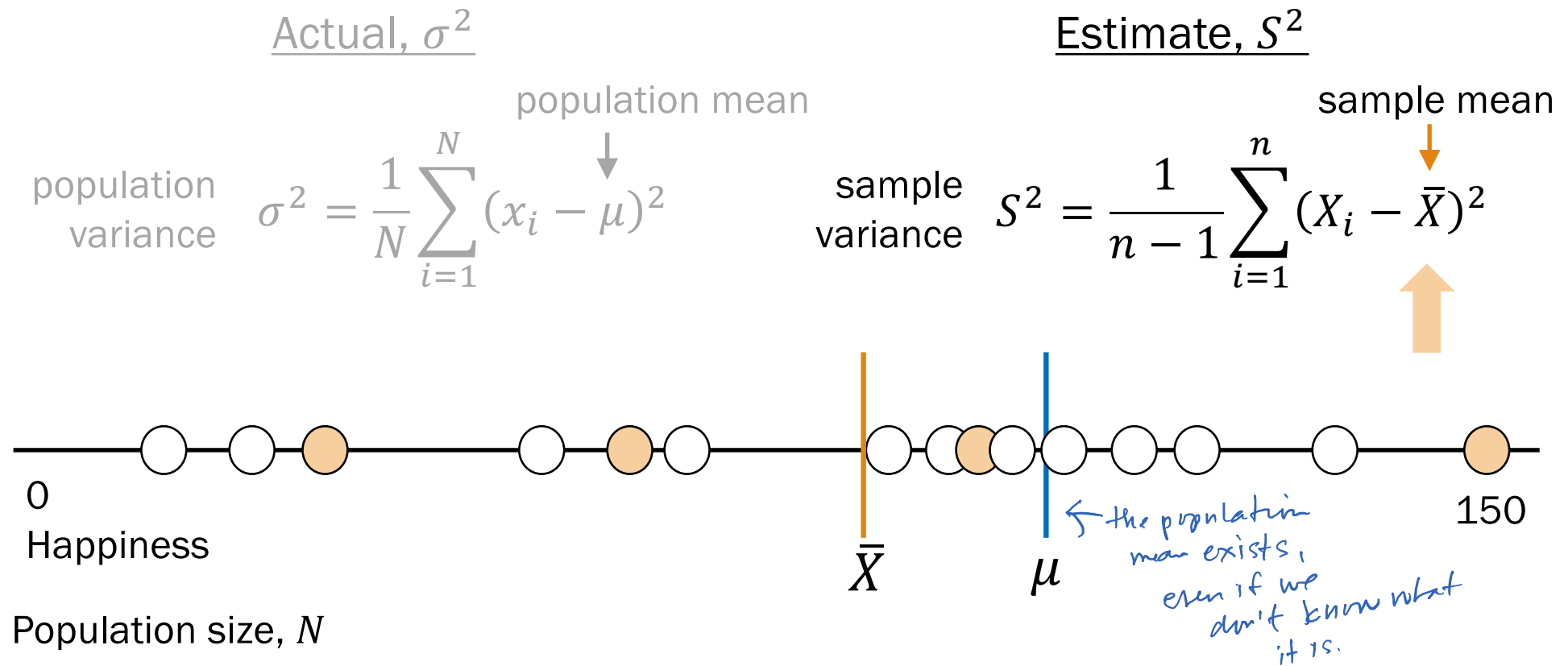
sample variance

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

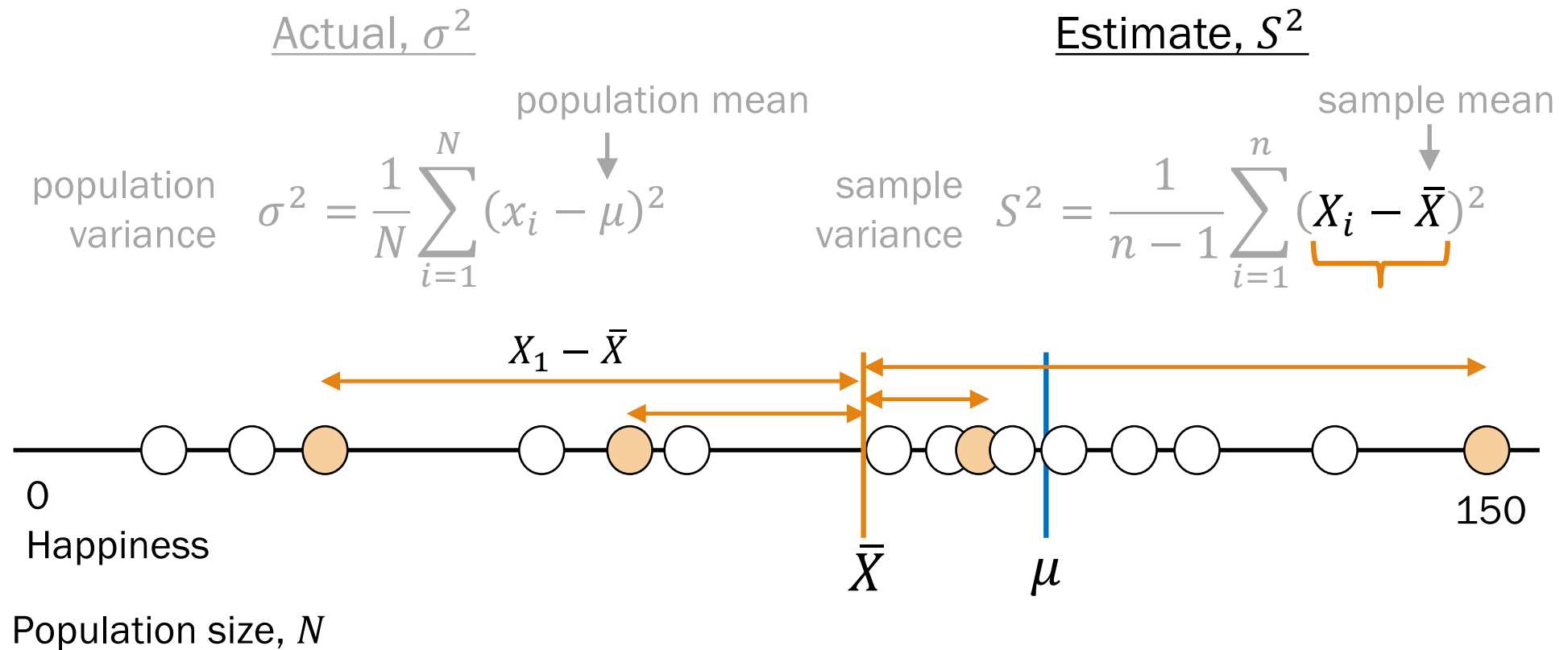
sample mean



Intuition about the sample variance, S^2



Intuition about the sample variance, S^2



Estimating the population variance



2. What is σ^2 , the **variance of happiness** of Bhutanese people?

If we only have a sample, $(X_1, X_2, \dots, X_n)$:

The best estimate of σ^2 is the **sample variance**:
$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

S^2 is an **unbiased estimator** of the population variance, σ^2 . $E[S^2] = \sigma^2$

Proof that S^2 is unbiased (just for reference)

$$E[S^2] = \sigma^2$$

$$E[S^2] = E\left[\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2\right] \Rightarrow (n-1)E[S^2] = E\left[\sum_{i=1}^n (X_i - \bar{X})^2\right]$$

work through this if you'd like to, or just ignore it

$$(n-1)E[S^2] = E\left[\sum_{i=1}^n ((X_i - \mu) + (\mu - \bar{X}))^2\right] \quad (\text{introduce } \mu - \mu)$$

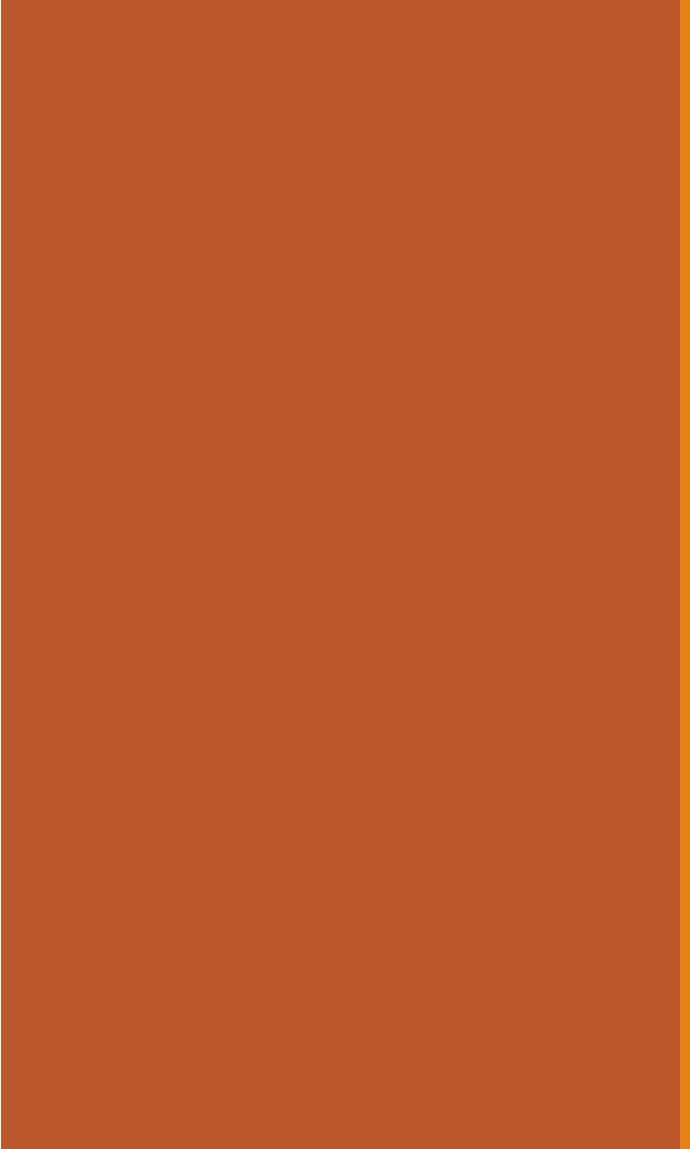
$$= E\left[\sum_{i=1}^n (X_i - \mu)^2 + \sum_{i=1}^n (\mu - \bar{X})^2 + 2 \sum_{i=1}^n (X_i - \mu)(\mu - \bar{X})\right]$$

$$= E\left[\sum_{i=1}^n (X_i - \mu)^2 + n(\mu - \bar{X})^2 - 2n(\mu - \bar{X})^2\right]$$

$$= E\left[\sum_{i=1}^n (X_i - \mu)^2 - n(\mu - \bar{X})^2\right] = \sum_{i=1}^n E[(X_i - \mu)^2] - nE[(\bar{X} - \mu)^2]$$

$$= n\sigma^2 - n\text{Var}(\bar{X}) = n\sigma^2 - n\frac{\sigma^2}{n} = n\sigma^2 - \sigma^2 = (n-1)\sigma^2$$

Therefore, $E[S^2] = \sigma^2$



Standard error

Estimating population statistics

A particular outcome

1. Collect a sample, $X_1, X_2, \dots, X_n$.

(72, 85, 79, 79, 91, 68, ..., 71)
 $n = 200$

2. Compute **sample mean**, $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$.

$\bar{X} = 83$

3. Compute sample deviation, $X_i - \bar{X}$.

(-11, 2, -4, -4, 8, -15, ..., -12)

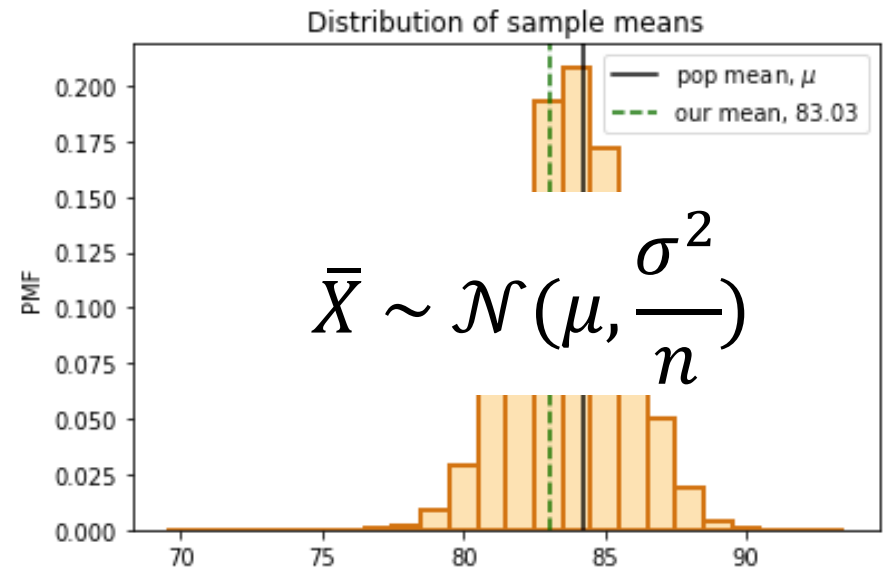
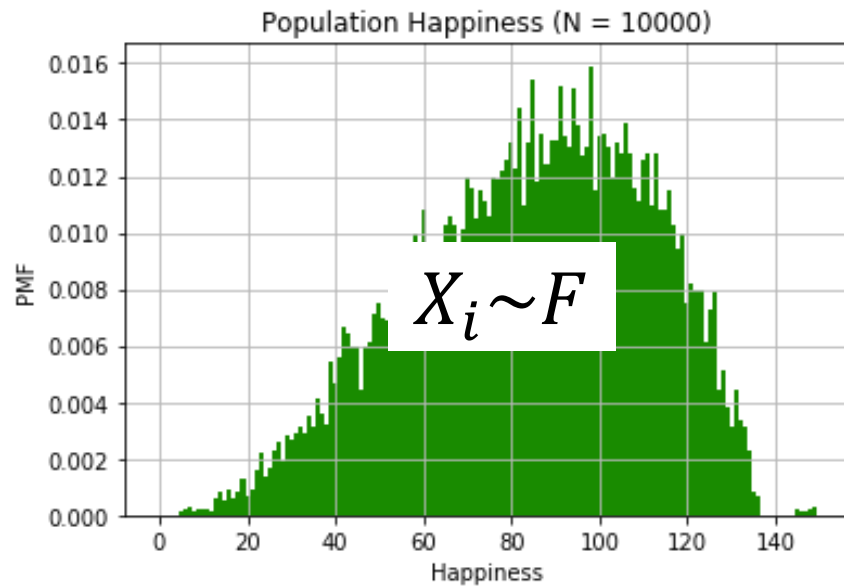
Handwritten annotations:
73 - 83
85 - 83
79 - 83
Arrows point from these calculations to the corresponding values in the sequence: -11, 2, -4, -4, 8, -15, ...

4. Compute **sample variance**, $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$.

$S^2 = 793$

How close are our estimates $\bar{X}$ and S^2 to the true μ and σ^2 ?

Sample mean



- $\text{Var}(\bar{X})$ is a measure of how close $\bar{X}$ is to μ .
- **How do we estimate $\text{Var}(\bar{X})$?**

if $\text{Var}(\bar{X})$ is large, then we're less confident that any single $\bar{X}$ is close to μ .

if, conversely, $\text{Var}(\bar{X})$ is small, then we're much more confident that $\bar{X}$ is close to μ .

How close is our estimate $\bar{X}$ to μ ?

$$E[\bar{X}] = \mu$$

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

We want to estimate this

$$\text{SD}(\bar{X}) = \sqrt{\text{Var}(\bar{X})} = \sqrt{\frac{\sigma^2}{n}}$$

Sadly, we don't know what σ^2 is, so we use S^2

def The **standard error** of the mean is an estimate of the standard deviation of $\bar{X}$.

Intuition:

- S^2 is an unbiased estimate of σ^2
- S^2/n is an unbiased estimate of $\sigma^2/n = \text{Var}(\bar{X})$
- $\sqrt{S^2/n}$ is an estimate of $\sqrt{\text{Var}(\bar{X})}$

it's biased, but it's the best we can do at the moment.

$$SE = \sqrt{\frac{S^2}{n}}$$

$$E[SE] < \text{SD}(\bar{X})$$

less than because of this bias.

More info on bias of standard error: [wikipedia](https://en.wikipedia.org/wiki/Bias_of_standard_error)

Standard error of the mean

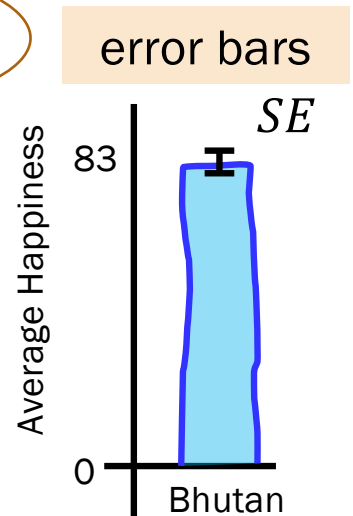
1. Mean happiness:

Claim: The average happiness of Bhutan is 83, with a standard error of 1.99.

Closed form: $SE = \sqrt{\frac{S^2}{n}}$

this is our estimate of how close we are to μ .

this is our best estimate of μ



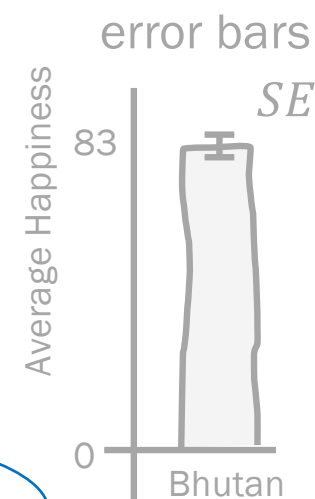
These 2 statistics give a sense of how $\bar{X}$ —that is, the sample mean random variable—behaves.

Standard error of variance?

1. Mean happiness:

Claim: The average happiness of Bhutan is 83, with a standard error of 1.99.

Closed form: $SE = \sqrt{\frac{S^2}{n}}$



2. Variance of happiness:

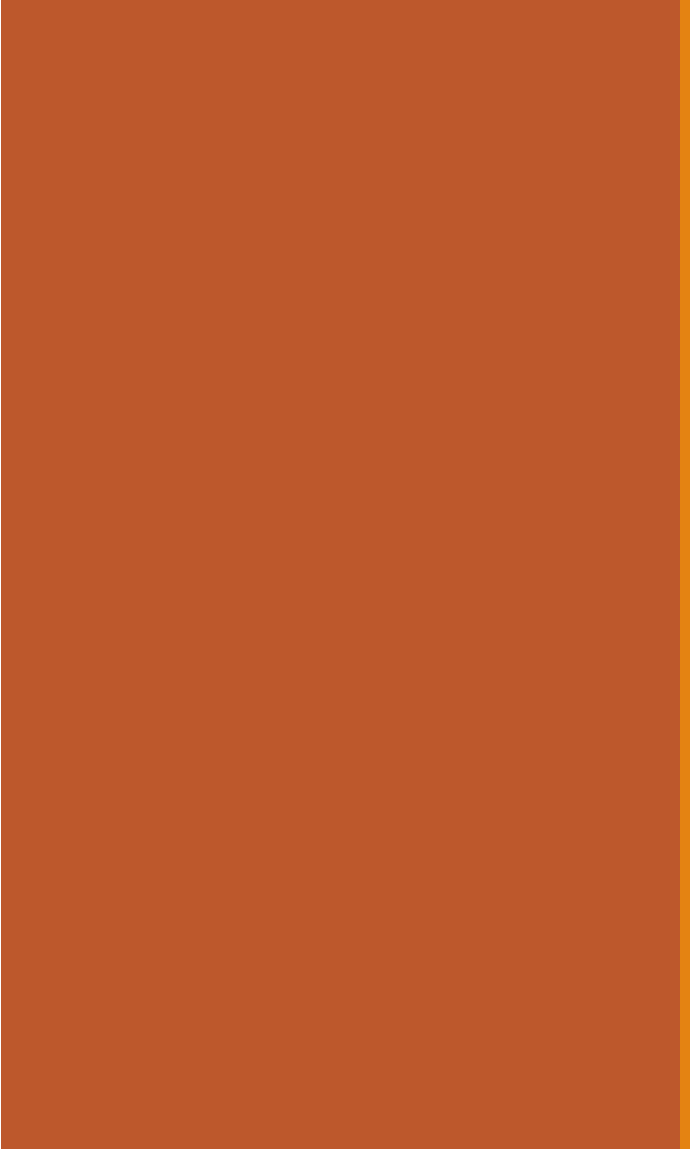
Claim: The variance of happiness of Bhutan is 793.

Closed form: Not covered in CS109

But how close are we?



Up next: Compute statistics with code!



Bootstrap: Sample mean

Bootstrap

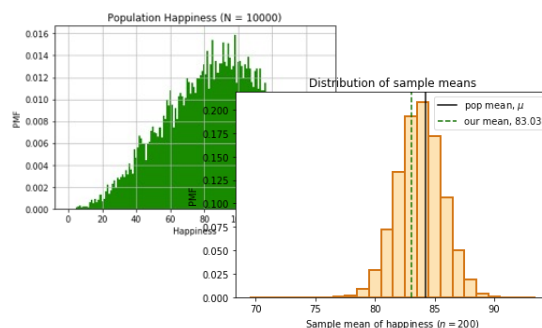
The Bootstrap:

Probability for Computer Scientists

Computing statistic of sample mean

What is the standard deviation of the sample mean $\bar{X}$? (sample size $n = 200$)

Population
distribution
(we don't have this)



$$\frac{\sigma}{\sqrt{n}} = 1.886$$

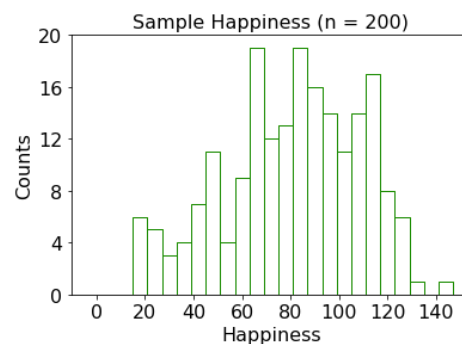
*these are the types
of things we can
compute if we have
the full
distribution*

Exact statistic
(we don't have this)

$$1.869$$

Simulated statistic
(we don't have this)

Sample
distribution
(we **do** have this)



$$SE = \frac{S}{\sqrt{n}} = 1.992$$

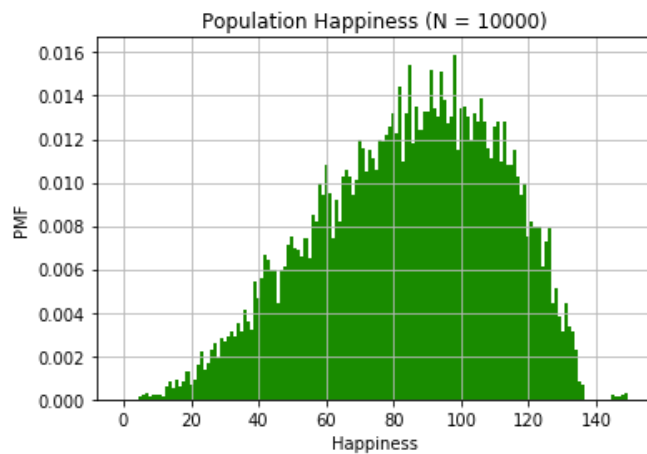
???

Estimated statistic,
by formula,
standard error

Simulated
estimated statistic

Note: We don't have access to the population.
But Doris is sharing the exact statistic with you.

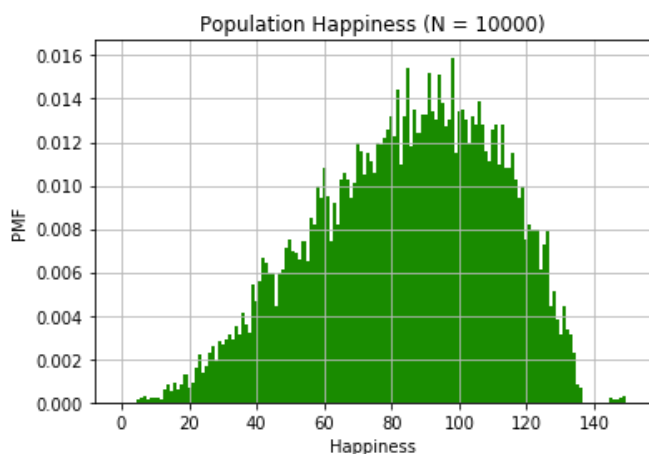
Bootstrap insight 1: Estimate the true distribution



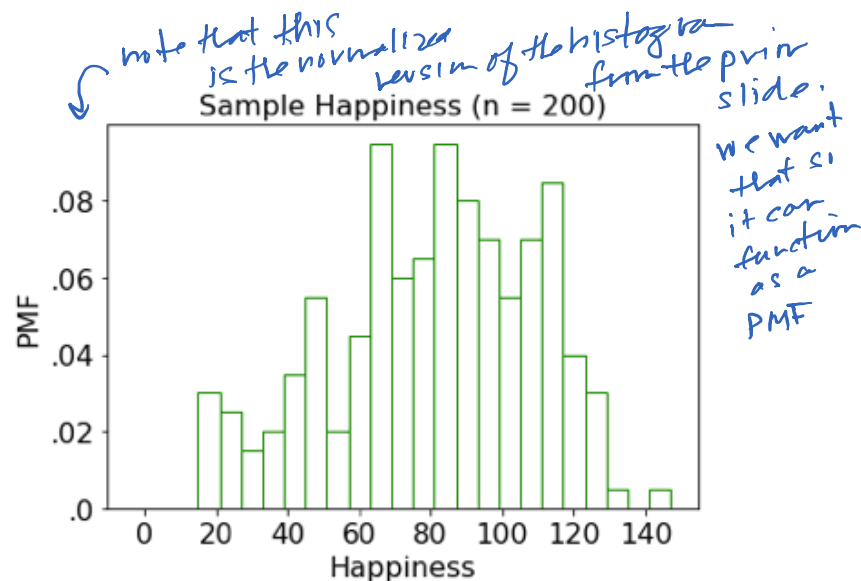
$\approx$

Bootstrap insight 1: Estimate the true distribution

You can estimate the PMF of the underlying distribution, using your sample.*



$\approx$



The underlying
distribution



$$F \approx \hat{F}$$



the sample distribution
(aka the histogram of
your sample data)

*This is just a histogram of your data!

John, Chris Piech, Mehran Sahami, and Jerry Cain, CS109, Spring 2024

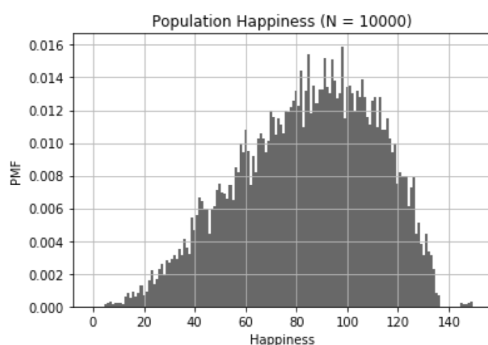
Bootstrap insight 2: Simulate a distribution

Approximate the procedure of simulating a distribution of a statistic, e.g., $\bar{X}$.

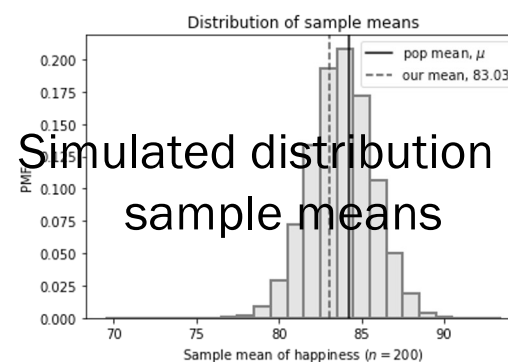
Population
distribution
(we don't have this)



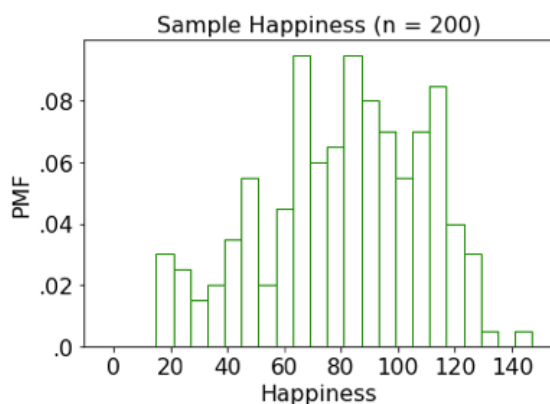
Sample
distribution
(we do have this)



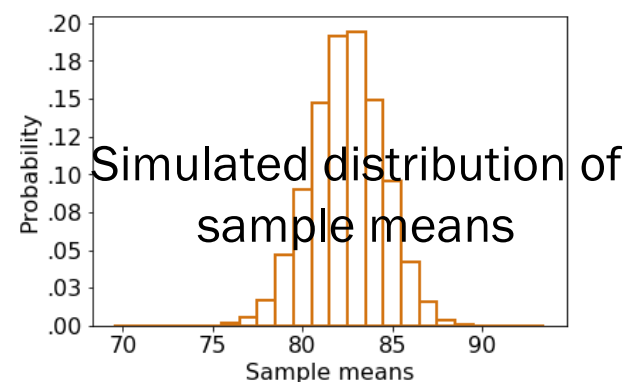
Distribution
of $\bar{X}$



Simulated distribution of
sample means



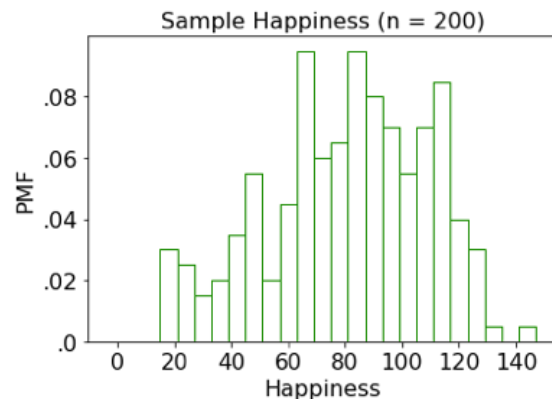
Bootstrap
means



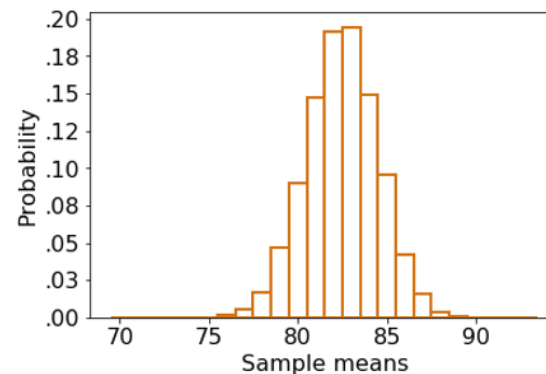
Simulated distribution of
sample means

Bootstrapped sample means

`means` = [84.7, 83.9, 80.6, 79.8, 90.3, ..., 85.2]



Estimate the true PMF
using our "PMF" (histogram)
of our sample.



...generate a whole
bunch of sample means
using this estimated distribution...

`np.std(means)`
2.003

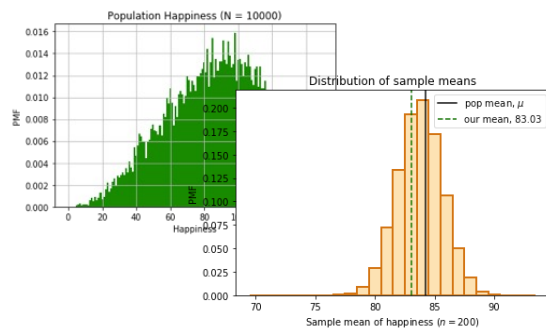
...and compute the
standard deviation
of this distribution.

*by brute force
computation.*

Computing statistic of sample mean

What is the standard deviation of the sample mean $\bar{X}$? (sample size $n = 200$)

Population
distribution
(we don't have this)



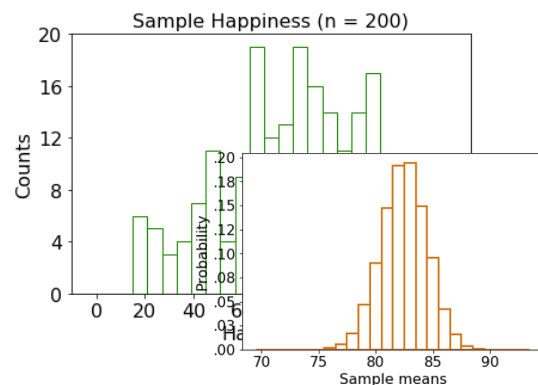
$$\frac{\sigma}{\sqrt{n}} = 1.886$$

Exact statistic
(we don't have this)

1.869

Simulated statistic
(we don't have this)

Sample
distribution
(we do have this)



$$SE = \frac{S}{\sqrt{n}} = 1.992$$

Estimated statistic,
by formula,
standard error

2.003

Simulated estimated
statistic, **bootstrapped
standard error**

Bootstrap algorithm

Bootstrap Algorithm (sample):

1. Estimate the **PMF** using the sample
2. Repeat **10,000** times:
 - a. Resample **sample.size()** from PMF
 - b. Recalculate the **sample mean** on the resample
3. You now have a **distribution of your sample mean**

We sample with replacement, because the sample operates as a PMF.

What is the distribution of your **sample mean**?

We'll talk about this algorithm in detail with a demo!

Bootstrap algorithm

Bootstrap Algorithm (sample):

1. Estimate the **PMF** using the sample
2. Repeat **10,000** times:
 - a. Resample **sample.size()** from PMF
 - b. Recalculate the **statistic** on the resample
3. You now have a **distribution of your statistic**

What is the distribution of your **statistic**?

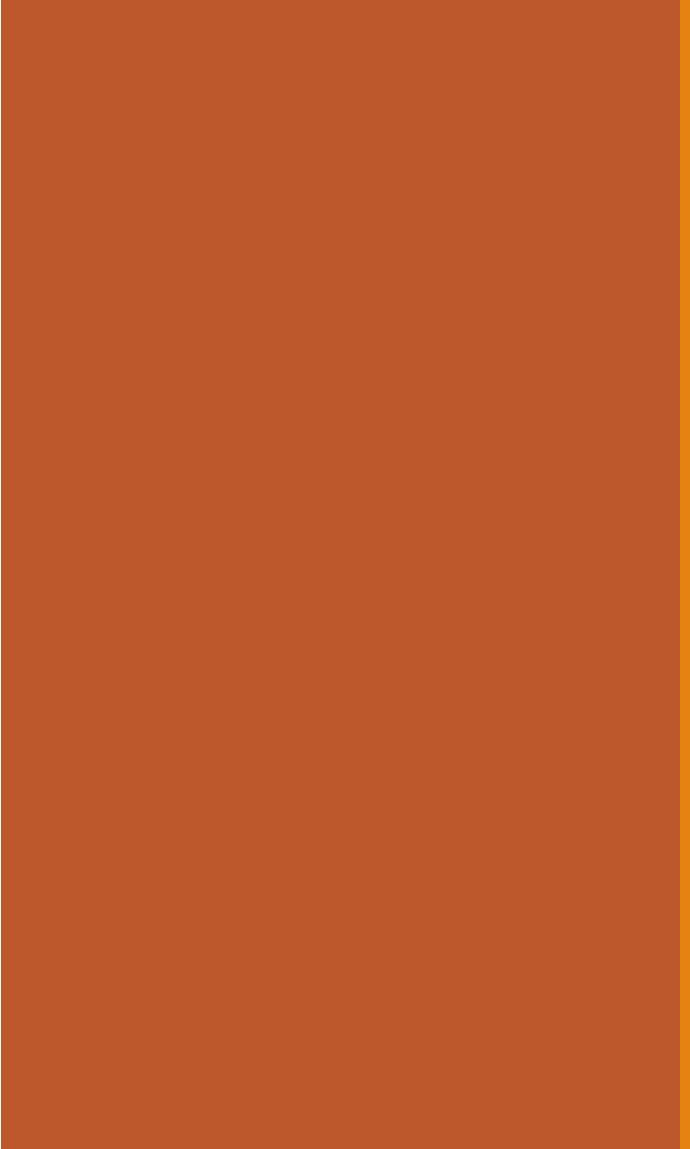
Bootstrapped sample variance

Bootstrap Algorithm (sample):

1. Estimate the **PMF** using the sample
2. Repeat **10,000** times:
 - a. Resample **sample.size()** from PMF
 - b. Recalculate the **sample variance** on the resample
3. You now have a **distribution of your sample variance**

What is the distribution of your **sample variance**?

Even without closed forms,
we estimate statistics of the sample variance with bootstrapping!



Bootstrap: Sample variance

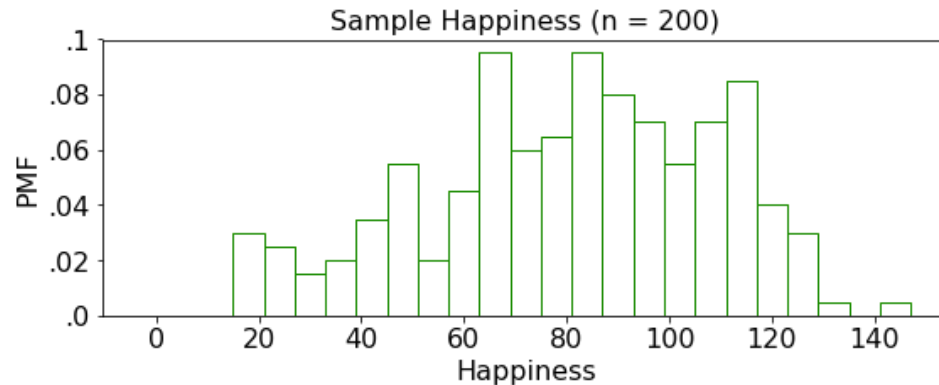
Bootstrapped sample variance

Bootstrap Algorithm (sample):

1. Estimate the **PMF** using the sample
2. Repeat **10,000** times:
 - a. Resample **sample.size()** from PMF
 - b. Recalculate the **sample variance** on the resample
3. You now have a **distribution of your sample variance**

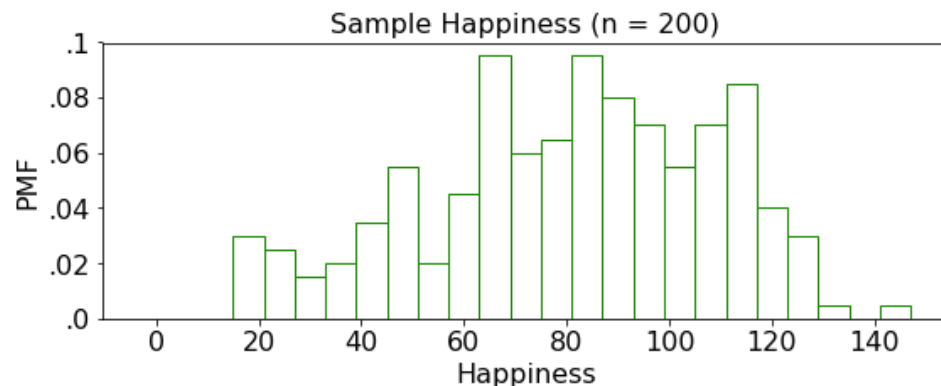
Goal What is the distribution of your **sample variance**?

Bootstrapped variance



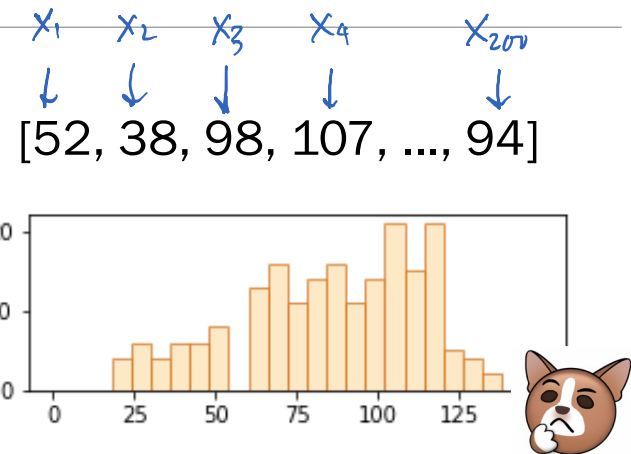
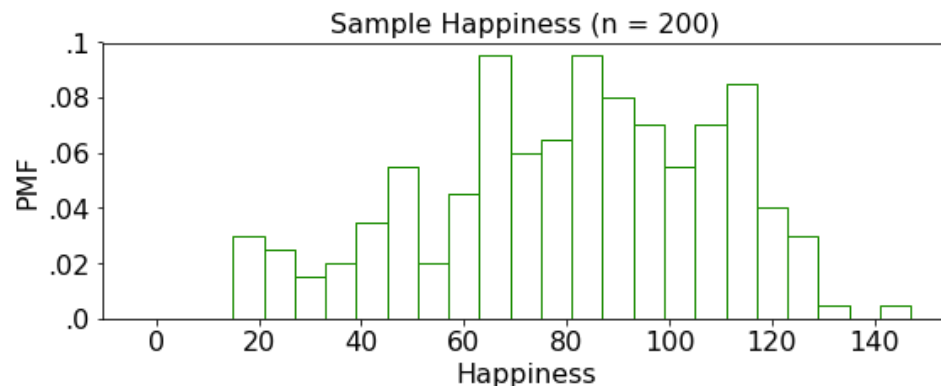
- ➔ 1. Estimate the **PMF** using the sample
2. Repeat **10,000** times:
 - a. Resample `sample.size()` from PMF
 - b. Recalculate the **sample variance** on the resample
3. You now have a **distribution of your sample variance**

Bootstrapped variance



1. Estimate the **PMF** using the sample
- ➔ 2. Repeat **10,000** times:
 - a. Resample **sample.size()** from PMF
 - b. Recalculate the **sample variance** on the resample
3. You now have a **distribution of your sample variance**

Bootstrapped variance



1. Estimate the PMF using the sample
2. Repeat **10,000** times:
 - a. Resample `sample.size()` from PMF
 - b. Recalculate the **sample variance** on the resample
3. You now have a **distribution of your**

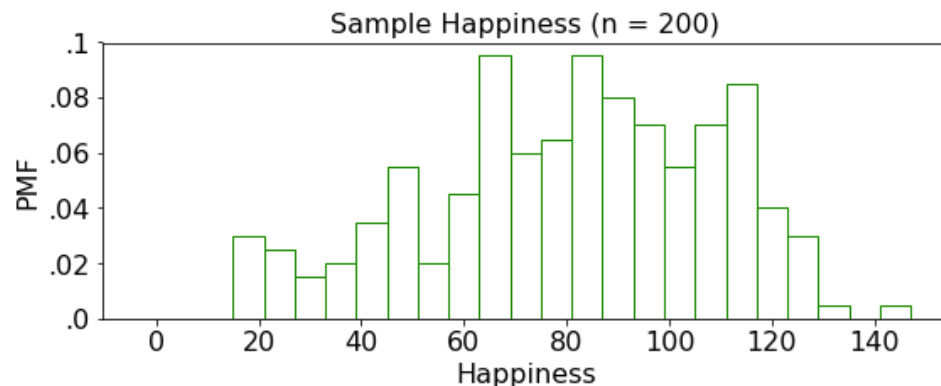


Why are these samples **different**?

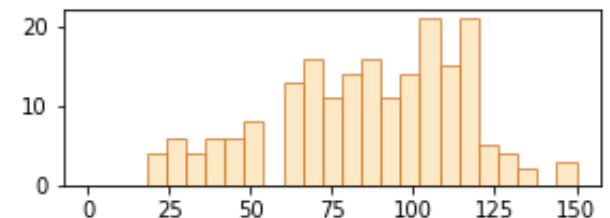
because we sample with replacement :)

This resampled sample is generated with **replacement**.

Bootstrapped variance



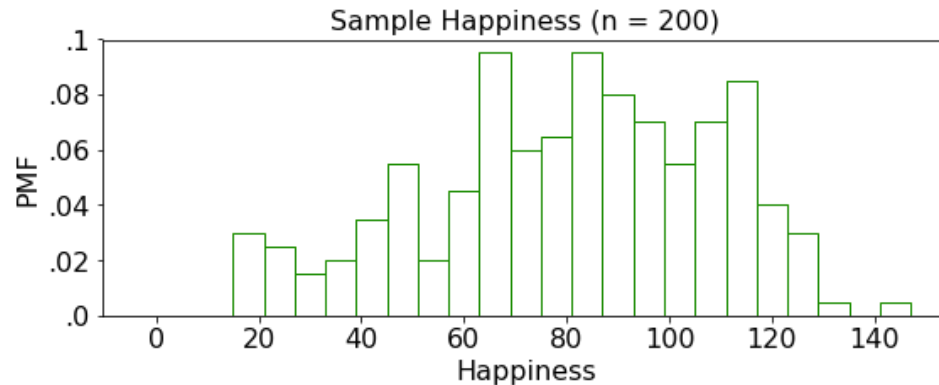
[52, 38, 98, 107, ..., 94]



1. Estimate the **PMF** using the sample
2. Repeat **10,000** times:
 - a. Resample **sample.size()** from PMF
 - ➡ b. Recalculate the **sample variance** on the resample
3. You now have a **distribution of your sample variance**

variances = [827.4]

Bootstrapped variance



1. Estimate the **PMF** using the sample



2. Repeat **10,000** times:

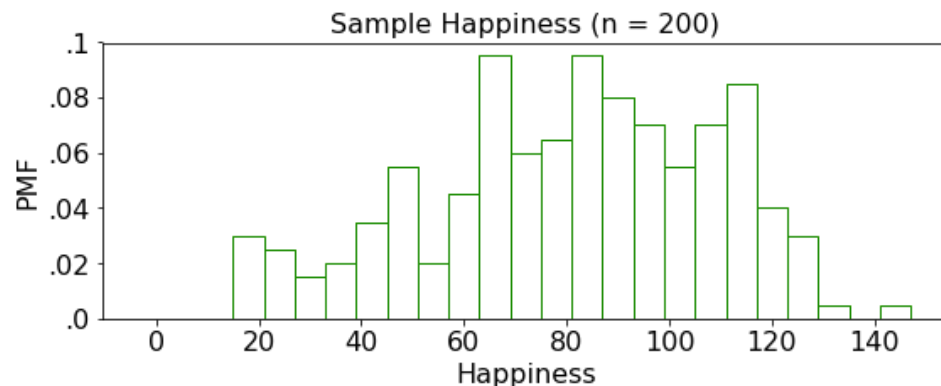
a. Resample **sample.size()** from PMF

b. Recalculate the **sample variance** on the resample

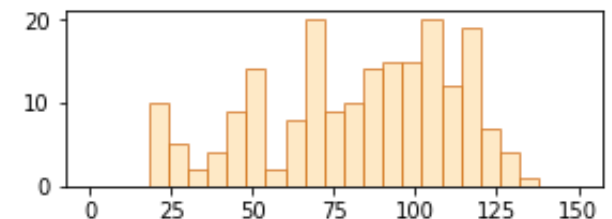
3. You now have a **distribution of your sample variance**

variances = [827.4]

Bootstrapped variance



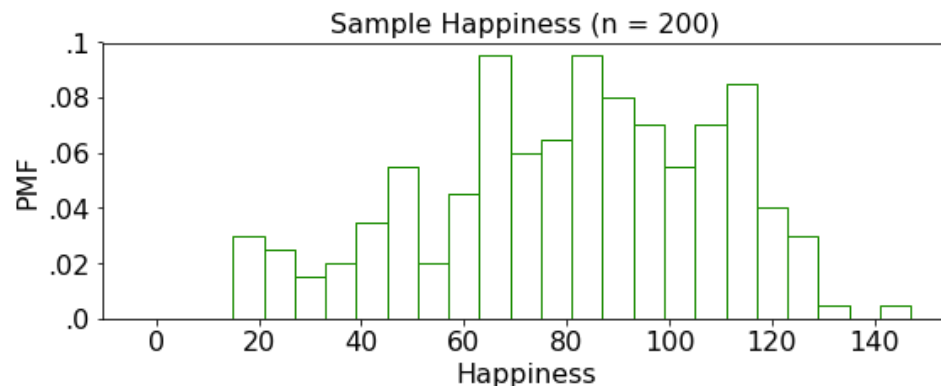
[116, 76, 132, 85, ..., 78]



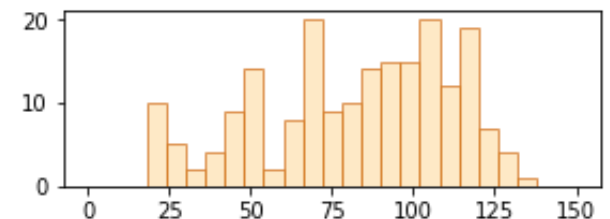
1. Estimate the **PMF** using the sample
2. Repeat **10,000** times:
 - a. Resample `sample.size()` from PMF
 - b. Recalculate the **sample variance** on the resample
3. You now have a **distribution of your sample variance**

variances = [827.4]

Bootstrapped variance



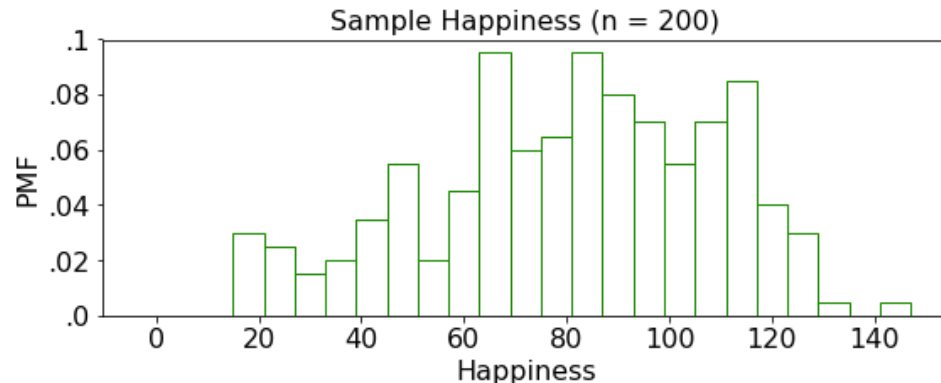
[116, 76, 132, 85, ..., 78]



1. Estimate the **PMF** using the sample
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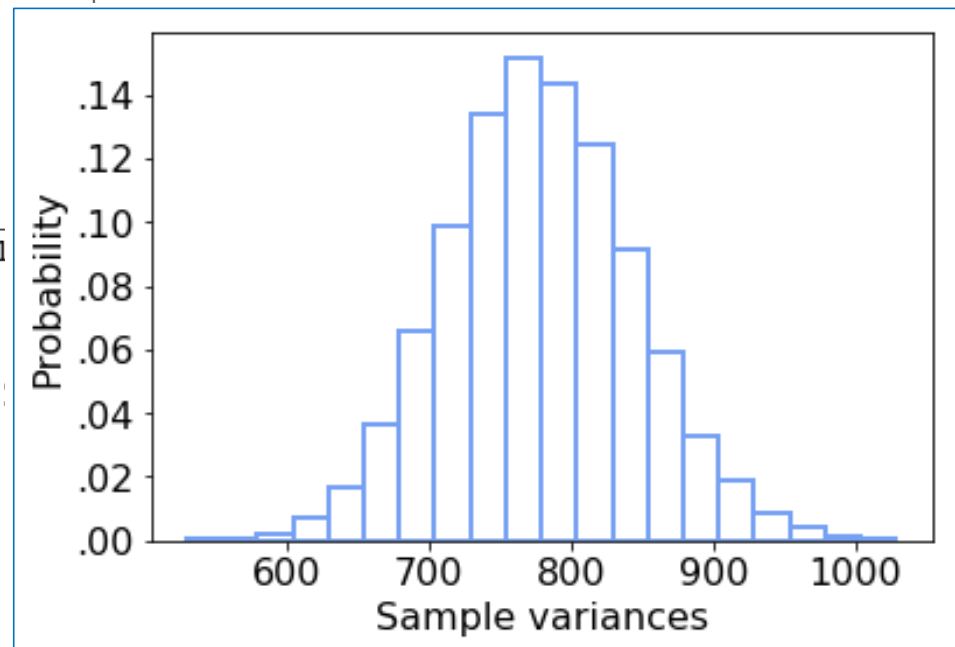
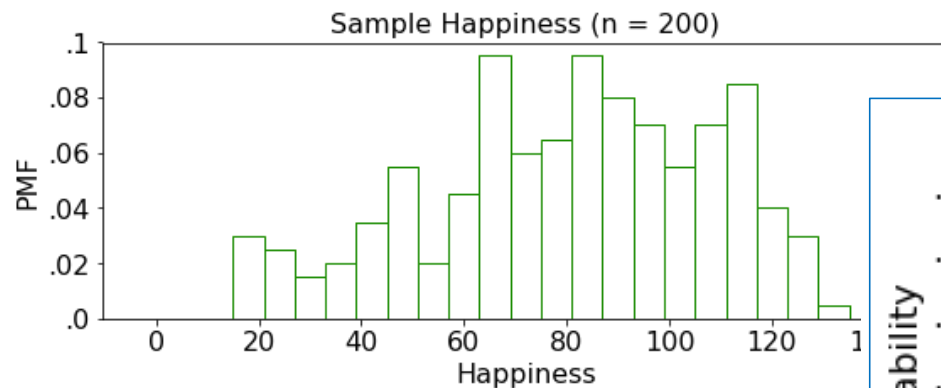
variances = [827.4, 846.1]

Bootstrapped variance



1. Estimate the **PMF** using the sample
 2. Repeat **10,000** times:
 - a. Resample **sample.size()** from PMF
 - b. Recalculate the **sample variance** on the resample
 3. You now have a **distribution of your sample variance**
- variances** = [827.4, 846.1]

Bootstrapped variance



1. Estimate the **PMF** using the
2. Repeat **10,000** times:
 - a. Resample **sample.size()**
 - b. Recalculate the **sample**

➡ 3. You now have a **distribution of your sample variance**

variances = [827.4, 846.1, 726.0, ..., 860.7]

Bootstrapped variance

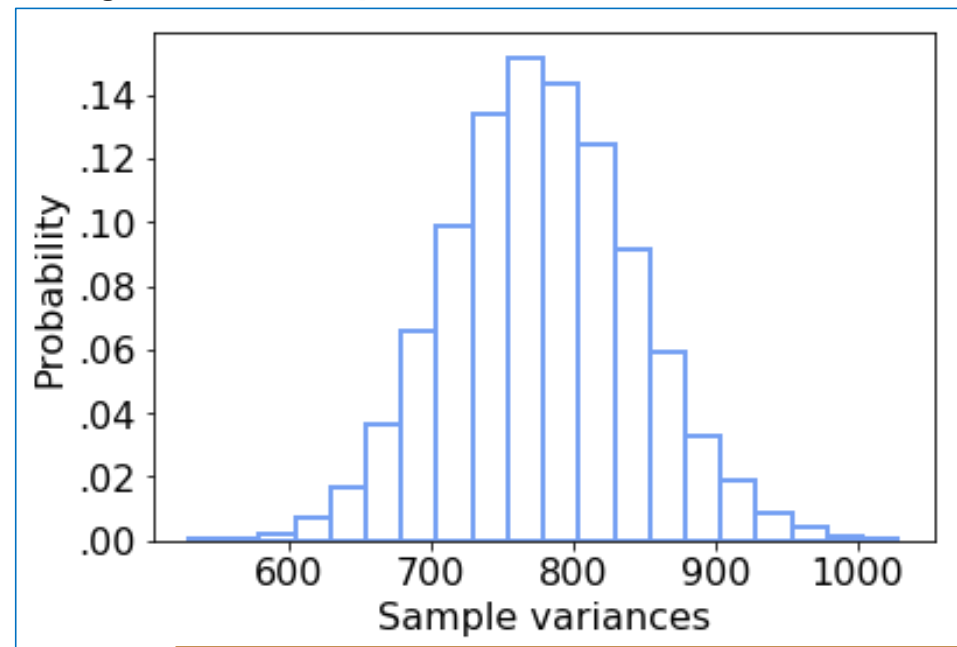
3. You now have a **distribution of your sample variance**

```
variances = [827.4,  
             846.1, 726.0, ...,  
             860.7]
```

What is the bootstrapped standard error?

```
np.std(variances)
```

Bootstrapped standard error: 66.16



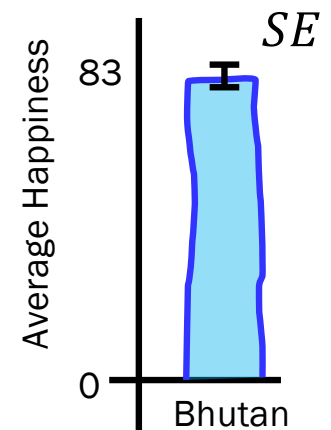
- Simulate a distribution of sample variances
- Compute standard deviation

Standard error

1. Mean happiness:

Claim: The average happiness of Bhutan is 83, with a standard error of 1.99.

Closed form: $SE = \sqrt{\frac{S^2}{n}}$



S^2 is our best estimate of σ^2

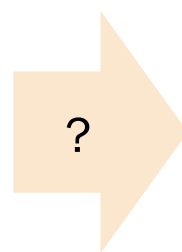
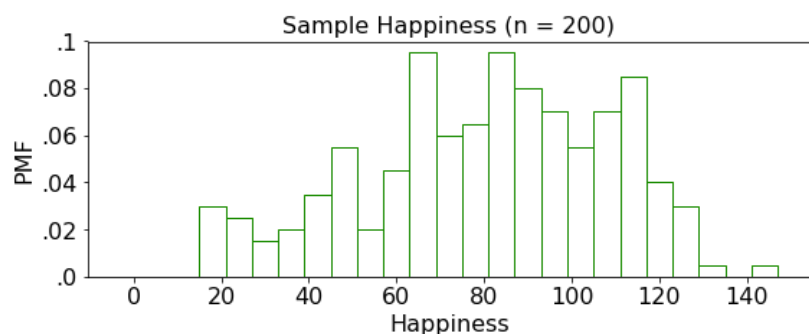
2. Variance of happiness:

Claim: The variance of happiness of Bhutan is 793, with a **bootstrapped standard error of 66.16**.

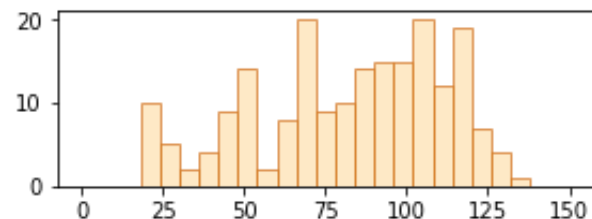
this is how confident we are

Algorithm in practice: Resampling

1. Estimate the **PMF** using the sample
2. Repeat **10,000** times:
 - a. Resample **sample.size()** from PMF
 - b. Recalculate the **statistic** on the resample
3. You now have a **distribution of your statistic**



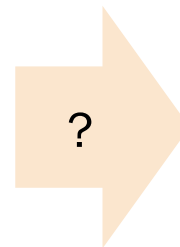
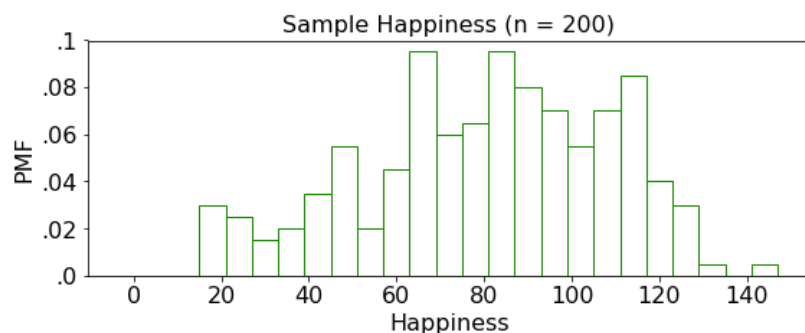
[116, 76, 132, 85, ..., 78]



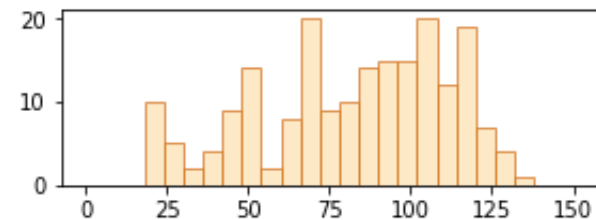
$$P(X = k) = \frac{\text{\# values in sample equal to } k}{n}$$

Algorithm in practice: Resampling

```
def resample(sample, n):  
    # estimate the PMF using the sample  
    # draw n new samples from the PMF  
    return np.random.choice(sample, n, replace = True)
```



[116, 76, 132, 85, ..., 78]



$$P(X = k) = \frac{\text{\# values in sample equal to } k}{n}$$

This resampled sample is generated with **replacement**.

To the code!

Bootstrap provides a way to calculate probabilities of statistics using code.

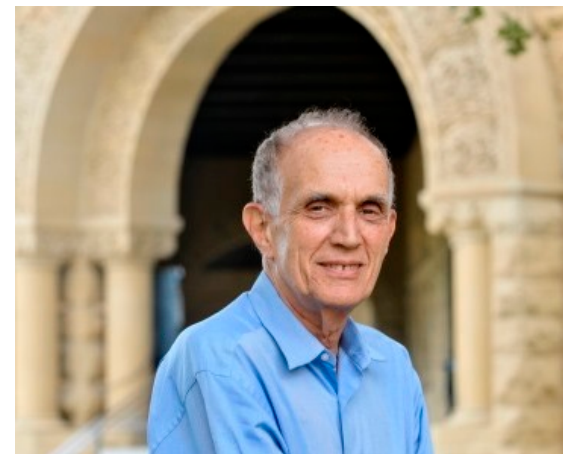
Bootstrapping works for any statistic*

*as long as your sample is iid and the underlying distribution does not have a long tail

Google colab notebook [link](#)

Bradley Efron

- Invented bootstrapping in 1979
- Still a professor at Stanford
- Won a National Science Medal



Inventor of Efron's dice: 4 dice A, B, C, D where:

$$P(A > B) = P(B > C) = P(C > D) = P(D > A) = \frac{2}{3}$$

